2.6. Electromagnets, induction & inductance

`em17 - Thomson`

Where does the energy come from?

New concepts:

- magnetic flux & induction
- electromagnets & transformers
- alternating current (AC)

Interaction of coil & magnet

- previous lecture we used **constant** magnetic field to understand:
    1. electric current produces magnetic field
    2. magnetic field exerts a force on electric currents/moving electric particles

What happens when we change the magnetic field over time?

`em18 - magnet drop`

- Joseph Henry (1797-1878) & Michael Faraday (1791-1867) described independently that **changing magnetic field induces an electric field**
- **concepts to understand**:
    - alternating voltage/current
    - electromagnetic induction
    - inductance & electromagnets

What influences the induced voltage ? (1/2)

`em12 - flux`

Impact of geometry and orientation

The flux strikes back: Magnetic flux

`sim mag flux`

- magnetic flux Φ_B defined as:

$$\Phi_B = \int \vec{B}\, d\vec{A}$$

- here $\vec{A}$ is the vector normal to the area A
- in contrast to the electric flux Φ_E, "regular" integral instead of closed surface integral
- for a **uniform field** simplifies to:

$$\Phi_B = B_\perp A = BA\cos\theta = \vec{B} \cdot \vec{A}$$

- measured in *weber* [Wb] = [T m^2]

What influences the induced voltage ? (2/2)

`em12 - flux`

Impact of change over time and number of loops

- **only changing magnetic fields induce a voltage**
- rate of change scales with induced voltage, i.e. $\frac{d\Phi_B}{dt}$
- for same $\frac{d\Phi_B}{dt}$, coil with more loops shows higher induced voltage

Faraday’s law of induction

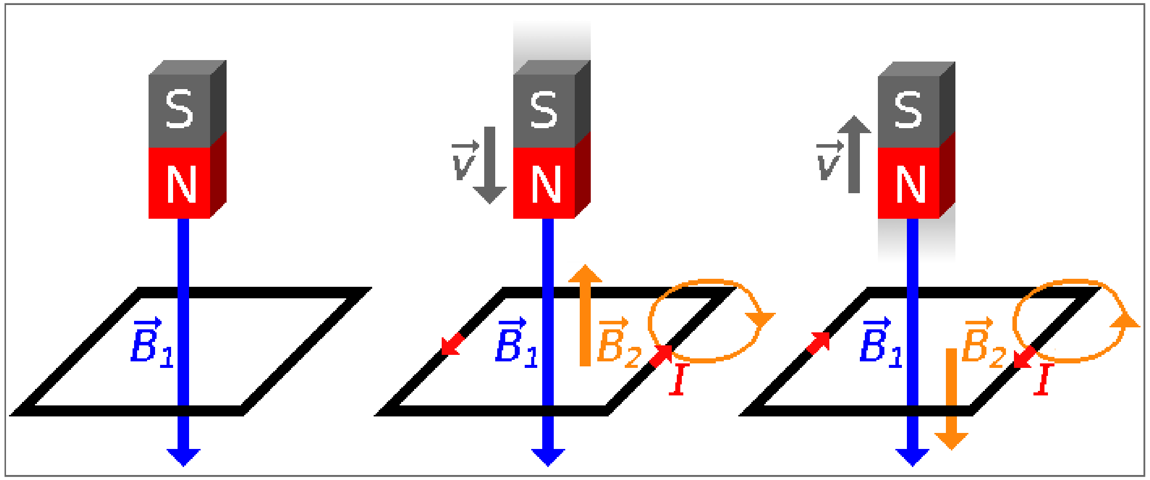
- $V_{ind} = -N\frac{d\Phi_B}{dt}$
- with:
    - N - number of loops in coil
    - $\frac{d\Phi_B}{dt}$ - change in magnetic flux

Why is there a minus?

Lenz's law

`em14 - Rings`

A change in the magnetic flux through a conducting loop induces a voltage, so that the resulting current generates a magnetic field that opposes the change in magnetic flux.



*from **wikipedia** under **CC0 1.0 Universal**; image was edited*

E-field from electrostatics vs. electromagnetism

Electrostatics	**Electromagnetism**
static charges generate electric fields	changing magnetic flux induces electric fields
field lines start/end on charges	field lines can form closed loops
voltage between two points: $V_{AB} = -\int_A^B \vec{E} \cdot d\vec{l}$	induced voltage: $V_{ind} = -\frac{d\Phi_B}{dt}$
$\oint \vec{E} \cdot d\vec{l} = 0$	$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$
conservative field / path independent	non-conservative field / closed-loop work can be non-zero

- electrostatics:
    - after returning to the starting point, a charge has no net energy gain
    - positive and negative work cancel exactly

- changing magnetic fields:
  - induced electric field continuously pushes charges around the loop
  - force on the charge always points along the loop
  - after one full loop, charges can gain energy

Eddy currents

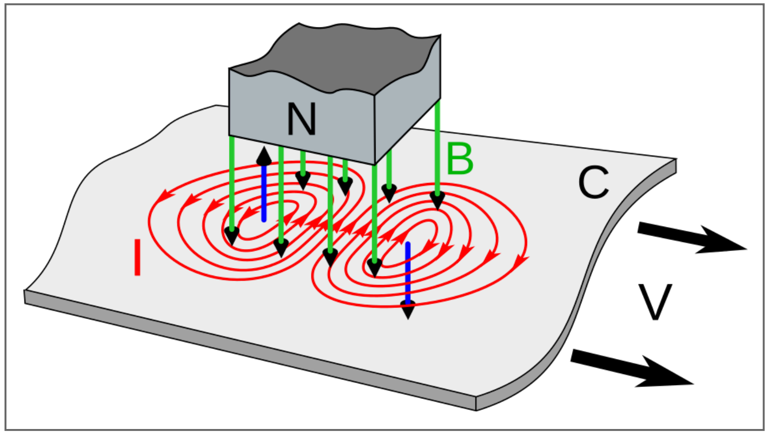
`em25 - Waltenhofen's pendulum`

- conductor (pendulum) moves through magnetic field generated by an electromagnet
- → changing magnetic flux in conductor → induced eddy currents
- part leaving the magnetic field:
    - decrease in magnetic flux (number of field lines passing through)
    - induced currents generate magnetic fields opposing the decrease (Lenz's law)
    - resulting force pulls the conductor back toward the electromagnet
- part entering the magnetic field:
    - increase in magnetic flux (number of field lines passing through)
    - induced currents generate magnetic fields opposing the increase (Lenz's law)

- resulting force pushes the conductor away from the electromagnet

Eddy currents (cont’)

`em25 - Waltenhofen's pendulum`



from ***wikipedia*** *under* ***CC0 1.0 Universal***

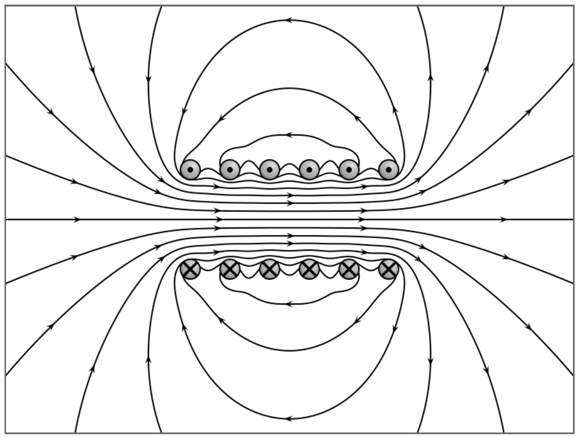
- **direction of B via RHR-1 by matching curl of fingers to induced current**

Electromagnets - solenoids

`em10 + simulation B-field solenoid`

- tightly wound helical coils of wire
- found in many devices: generators, loudspeakers, MRI scanners, and many more

- produce an almost uniform magnetic field inside the coil and weak field outside
- with loop density n, the field is: $B = \mu_0 \frac{N}{l} I = \mu_0 n I$



From induction to alternating current

- Faraday's law: **changing magnetic flux induces voltage**

$$V_{ind} = -\frac{d\Phi_B}{dt}$$

- → if magnetic flux changes periodically, **induced voltage also changes periodically**
- induced voltage therefore naturally alternates between positive and negative values

$$V(t) = V_0 \sin(\omega t)$$

- this is called **alternating current (AC)**

Primer on alternating current (AC)

- AC = periodically changing voltage and current
- sinusoidal AC is the most important case:

$$V(t) = V_0 \sin(2\pi f t) = V_0 \sin(\omega t)$$

$$I(t) = I_0 \sin(\omega t)$$

- angular frequency:

$$\omega = 2\pi f$$

- standard power grid:
    - most of the world: 50 Hz
    - north America: 60 Hz

Using Root-Mean-Squared (RMS) values

- often RMS values are used in AC:

$$V_{RMS} = \sqrt{\overline{V(t)^2}} = \sqrt{\overline{V_0^2 \sin^2(\omega t)}}$$

- with $\overline{\sin^2(\omega t)} = \frac{1}{2}$

$$V_{RMS} = \frac{V_0}{\sqrt{2}}$$

$$I_{RMS} = \frac{I_0}{\sqrt{2}}$$

- for example, a 220V outlet has a peak voltage $V_0 \approx 311$ V

Power in AC circuits

- power is $P(t) = I(t)V(t)$
- for sinusoidal current/voltage, average power over a period is **non-zero**:

$$\overline{P} = I_0 \sin(\omega t) V_0 \sin(\omega t) = I_0 V_0 \sin^2(\omega t)$$

- for one period $\sin^2(\omega t) = \frac{1}{2}$, $I_{RMS} = \frac{I_0}{\sqrt{2}}$ and $V_{RMS} = \frac{V_0}{\sqrt{2}}$

$$\overline{P} = \sqrt{2} I_{RMS} \sqrt{2} V_{RMS} \frac{1}{2} = I_{RMS} V_{RMS}$$

Transformers

`em46`

- transformer consists of a *primary* and *secondary* coil
- two coils **do not** form a closed circuit, but only their proximity is essential for the transformer to work

What influences the voltage in the secondary coil?

Transformers - What influences the voltage in the secondary coil?

- iron core (geometry, layered, etc.)
- ratio of loops:
  - without energy loss, the induced voltages in both coils are:

$$V_2 = N_2 \frac{d\Phi_B}{dt} \quad \& \quad V_1 = N_1 \frac{d\Phi_B}{dt}$$

  - taking the voltage ratio & using conservation of power, $P_1 = P_2$, we get:

$$\frac{V_2}{V_1} = \frac{I_1}{I_2} = \frac{N_2}{N_1}$$

- depending on the ratio of the loop numbers:
  - step-up transformer: For $N_2 > N_1$ the voltage V_2 will be larger than V_1

- step-down transformer: For $N_2 < N_1$ the voltage V_2 will be smaller than V_1

Mutual inductance

- relates the induced voltage in one coil to the rate of change of current in a neighboring coil
- defined as proportionality constant:

$$M_{21} = \frac{N_2 \Phi_{21}}{I_1}$$

- the induced voltage in one coil can directly be related to change in current in the other coil:

$$V_2 = -N_2 \frac{d\Phi_{21}}{dt} = -M_{21} \frac{dI_1}{dt} \quad \& \quad V_1 = -M_{12} \frac{dI_2}{dt}$$

- mutual inductance only depends on geometric variables, i.e. the spacing between the coils and their individual shape, size, and number of loops:

$$M = M_{21} = M_{12}$$

- **inductance** is expressed in **units of henry** [H] = [V s / A] = [Ω s]

Self-inductance & inductors

YouTube - James Bond

`em13`

- even for a **single coil**, we find the phenomenon of inductance, a.k.a **self-inductance**
- **changing current** passes through the coil **induces voltage in coil itself** with **opposed direction**
- defined as proportionality constant (depends on coil geometry):

$$L = \frac{N\Phi_B}{I}$$

$$V_{ind} = -N\frac{d\Phi_B}{dt} = -L\frac{dI}{dt}$$

- often referred simply to as inductance and measured in henry [H].

- **inductors are fundamental circuit elements alongside resistors and capacitors**

Revisit inital experiment & energy stored in the magnetic field

`em17 - Thomson`

- power supplied to an inductor is:

$$P = I V_{ind} = I L \frac{dI}{dt}$$

- the work done to build up the current is:

$$dW = P\,dt = IL\frac{dI}{dt}dt = ILdI$$

$$W = \int_0^I I L\,dI = \frac{1}{2}LI^2$$

- energy stored in the magnetic field is: $U = \frac{1}{2}LI^2$

- analogous to the energy stored in a capacitor: $U = \frac{1}{2}CV^2$